

IN410 set of Exercises (supplement)

Exercise 1:

Using the extended Euclidean algorithm, find the multiplicative inverse of:

- a. $135 \bmod 61$,
- b. $7465 \bmod 2464$, and
- c. $42828 \bmod 6407$.

Answer key:

- a. 3239
- b. $\gcd(40902, 24240) = 34 \neq 1$, so there is no multiplicative inverse.
- c. 550

Exercise 2:

It can be shown (see any book on number theory) that if $\gcd(m, n) = 1$, then $f(mn) = f(m)f(n)$. Using this property, the property developed in the preceding problem, and the property that $f(p) = p - 1$ for p prime, it is straightforward to determine the value of $f(n)$ for any n . Determine the following:

- a. $f(29)$ b. $f(51)$ c. $f(455)$ d. $f(616)$

Answer key:

- a. $f(41) = 40$, because 41 is prime
- b. $f(27) = f(3^3) = 3^3 - 3^2 = 27 - 9 = 18$
- d. $f(231) = f(3) \times f(7) \times f(11) = 2 \times 6 \times 10 = 120$
- c. $f(440) = f(23) \times f(5) \times f(11) = (2^3 - 2^2) \times 4 \times 10 = 160$

Exercise 3:

In one of his cases, Sherlock Holmes was confronted with the following message.

534 C2 13 127 36 31 4 17 21 41
DOUGLAS 109 293 5 37 BIRLSTONE
26 BIRLSTONE 9 127 171

Although Watson was puzzled, Holmes was able immediately to deduce the type of cipher. Can you?

Answer key:

The cipher refers to the words in the page of a book. The first entry, 534, refers to page 534. The second entry, C2, refers to column two. The remaining numbers are words in that column. The names DOUGLAS and BIRLSTONE are simply words that do not appear on that page.

Elementary! (from The Valley of Fear, by Sir Arthur Conan Doyle)

- a. Using this Playfair matrix:

J/K	C	D	E	F
U	N	P	Q	S
Z	V	W	X	Y
R	A	L	G	O
B	I	T	H	M

Encrypt this message:

I only regret that I have but one life to give for my country.

Note: This message is by Nathan Hale, a soldier in the American Revolutionary War.

b. Repeat part (a) using the Playfair matrix by Constructing it with the key *algorithm*.

Answer key:

a. UZTBDLGZPNNWLGTGTUEROVLDBDUHFPERHWQSRZ

b. UZTBDLGZPNNWLGTGTUEROVLDBDUHFPERHWQSRZ

Exercise 4:

Encrypt the message “meet me at the usual place at ten rather than eight o clock”

using the Hill cipher with the key $\begin{pmatrix} 7 & 3 \\ 2 & 5 \end{pmatrix}$. Show your calculations and the result.

Answer key:

We need an even number of letters, so append a "q" to the end of the message. Then convert the letters into the corresponding alphabetic positions:

M	e	e	t	m	e	a	t	t	h	e	u	s	u	a	l
13	5	5	20	13	5	1	20	20	8	5	21	19	21	1	12
P	l	a	c	e	a	t	t	e	n	r	a	t	h	e	r
16	12	1	3	5	1	20	20	5	14	18	1	20	8	5	18
T	h	a	n	e	i	g	h	t	o	c	l	o	c	k	q
20	8	1	14	5	9	7	8	20	15	3	12	15	3	11	17

The calculations proceed two letters at a time. The first pair:

$$\begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = \begin{pmatrix} 9 & 4 \\ 5 & 7 \end{pmatrix} \begin{pmatrix} 13 \\ 5 \end{pmatrix} \bmod 26 = \begin{pmatrix} 137 \\ 100 \end{pmatrix} \bmod 26 = \begin{pmatrix} 7 \\ 22 \end{pmatrix}$$

The first two ciphertext characters are alphabetic positions 7 and 22, which correspond to GV. The complete ciphertext:

GVUIGVKODZYPUEHKJHUZWFFZFSJSDZMUDZMYCJQMFWWUQRKR

Exercise 5:

Using the Vigenère cipher, encrypt the word “explanation” using the word “leg”.

Answer key:

key: legleglegle

plaintext: explanation

ciphertext: PBVWETLXOZR

Exercise 6:

Determine the multiplicative inverse of $x^2 + 1$ in $GF(23)$ with $m(x) = x^3 + x - 1$.

Answer key:

$x^2 + 1$

Exercise 7:

In a public-key system using RSA, you intercept the ciphertext $C = 20$ sent to a user whose public key is $e = 13$, $n = 77$. What is the plaintext M ?

Answer key:

$M = 5$.

Exercise 8:

Alice and Bob use the Diffie–Hellman key exchange technique with a common prime $q = 23$ and a primitive root $a = 5$.

- If Bob has a public key $Y_B = 10$, what is Bob's private key X_B ?
- If Alice has a public key $Y_A = 8$, what is the shared key K with Bob?
- Show that 5 is a primitive root of 23.

Answer key:

a. $\phi(23) = 22$

$5^{22} = 1 \pmod{23}$

If you check 5^n for $n < 22$, you will find that none of the values is 1 mod 23.

b. 6, because $5^6 \pmod{23} = 10$

c. $K = 5^6 \pmod{23} = 8$

Exercise 9:

Suppose Alice and Bob use an Elgamal scheme with a common prime $q = 157$ and a primitive root $a = 5$.

- If Bob has public key $Y_B = 10$ and Alice chose the random integer $k = 3$, what is the ciphertext of $M = 9$?
- If Alice now chooses a different value of k so that the encoding of $M = 9$ is $C = (25, C_2)$, what is the integer C_2 ?

Answer key:

a. $(49, 57)$

b. $C_2 = 29$

Exercise 10:

you are using AES as symmetric encryption algorithm. Given the plaintext $\{0E0F0A0B0C0D09080706050403020100\}$ and the key $\{030202020202020302020202020202\}$:

- Show the original contents of State, displayed as a 4×4 matrix.
- Show the value of State after initial AddRoundKey.
- Show the value of State after SubBytes.
- Show the value of State after ShiftRows.
- Show the value of State after MixColumns.

Answer key:

a) Original contents of State

Plaintext bytes by columns:

- Col 0: 0E 0F 0A 0B
- Col 1: 0C 0D 09 08
- Col 2: 07 06 05 04
- Col 3: 03 02 01 00

So the State is:

$$\begin{bmatrix} 0E & 0C & 07 & 03 \\ 0F & 0D & 06 & 02 \\ 0A & 09 & 05 & 01 \\ 0B & 08 & 04 & 00 \end{bmatrix}$$

b) State after initial AddRoundKey

Key bytes by columns:

- Col 0: 03 02 02 02
- Col 1: 02 02 02 03
- Col 2: 02 02 02 02
- Col 3: 02 02 02 02

XOR each byte:

$$\begin{bmatrix} 0D & 0E & 05 & 01 \\ 0D & 0F & 04 & 00 \\ 08 & 0B & 07 & 03 \\ 09 & 0B & 06 & 02 \end{bmatrix}$$

c) State after SubBytes

Using the AES S-box:

- 0D \to D7
- 0E \to AB
- 05 \to 6B
- 01 \to 7C
- 0F \to 76
- 04 \to F2
- 00 \to 63
- 08 \to 30

- 0B \to 2B
- 07 \to C5
- 03 \to 7B
- 09 \to 01
- 06 \to 6F
- 02 \to 77

So:

$$\begin{bmatrix} D7 & AB & 6B & 7C \\ D7 & 76 & F2 & 63 \\ 30 & 2B & C5 & 7B \\ 01 & 2B & 6F & 77 \end{bmatrix}$$

d) State after ShiftRows

Shift each row left by its row number:

- Row 0: no shift
- Row 1: left by 1
- Row 2: left by 2
- Row 3: left by 3

$$\begin{bmatrix} D7 & AB & 6B & 7C \\ 76 & F2 & 63 & D7 \\ C5 & 7B & 30 & 2B \\ 77 & 01 & 2B & 6F \end{bmatrix}$$

e) State after MixColumns

Applying AES MixColumns to the first column gives:

For the first column after ShiftRows, we have:

$$\begin{bmatrix} D7 \\ 76 \\ C5 \\ 77 \end{bmatrix}$$

AES MixColumns multiplies this column by:

$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix}$$

in $GF(2^8)$.

So the new first column is:

$$\begin{bmatrix} (02 \cdot D7) \oplus (03 \cdot 76) \oplus C5 \oplus 77 \\ D7 \oplus (02 \cdot 76) \oplus (03 \cdot C5) \oplus 77 \\ D7 \oplus 76 \oplus (02 \cdot C5) \oplus (03 \cdot 77) \\ (03 \cdot D7) \oplus 76 \oplus C5 \oplus (02 \cdot 77) \end{bmatrix}$$

Repeating the same process for the remaining columns, you will obtain:

$$\begin{bmatrix} 9D & 3A & 68 & DE \\ 18 & D8 & D6 & DB \\ A9 & AC & 15 & 4C \\ 3F & 6D & B8 & A6 \end{bmatrix}$$

Exercise 11:

This problem provides a numerical example of encryption using a one-round version of DES.

We start with the same bit pattern for the key K and the plaintext, namely:

Hexadecimal notation: F E D C B A 9 8 7 6 5 4 3 2 1 0

Binary notation: 1111 1110 1101 1100 1011 1010 1001 1000 0111 0110 0101 0100 0011 0010 0001 0000

- Derive K_1 , the first-round subkey.
- Derive L_0, R_0 .
- Expand R_0 to get $E[R_0]$, where $E[\#]$ is the expansion function of Table C.1.
- Calculate $A = E[R_0] \oplus K_1$.
- Group the 48-bit result of (d) into sets of 6 bits and evaluate the corresponding S-box substitutions.
- Concatenate the results of (e) to get a 32-bit result, B.
- Apply the permutation to get $P(B)$.
- Calculate $R_1 = P(B) \oplus L_0$.
- Write down the ciphertext.

Answer key:

a) Derive K_1 , the first-round subkey

1) Apply PC-1 to the key

After PC-1, the 56-bit result splits into:

$$C_0 = 0F3355F$$

$$D_0 = 55330FF$$

2) Left shift each half by 1

Round 1 uses a 1-bit circular left shift:

$$\begin{aligned} C_1 &= 1E66ABE \\ D_1 &= AA661FE \end{aligned}$$

3) Apply PC-2

Thus the first-round 48-bit subkey is:

$$K_1 = \boxed{F4FD9864B65A}$$

b) Derive L_0, R_0

Apply the DES initial permutation IP to the plaintext FEDCBA9876543210.

$$IP(\text{FEDCBA9876543210}) = 33FF33000F550F55$$

Split into two 32-bit halves:

$$\begin{aligned} L_0 &= \boxed{33FF3300} \\ R_0 &= \boxed{0F550F55} \end{aligned}$$

c) Expand R_0 to get $E[R_0]$

Using the DES expansion table:

$$E(R_0) = \boxed{85EAAA85EAAA}$$

d) Calculate $A = E(R_0) \oplus K_1$

$$A = 85EAAA85EAAA \oplus F4FD9864B65A$$

$$\boxed{A = 711732E15CF0}$$

e) Group the 48-bit result into 6-bit groups and apply the S-boxes

$$A = 711732E15CF0$$

Binary form:

011100 010001 011100 110010 111000 010101 110011 110000

So the 8 groups are:

1. 011100
2. 010001
3. 011100
4. 110010
5. 111000
6. 010101
7. 110011
8. 110000

Applying S_1 to S_8 :

- $S_1(011100) = 0000$
- $S_2(010001) = 1100$
- $S_3(011100) = 0010$
- $S_4(110010) = 0001$
- $S_5(111000) = 0110$
- $S_6(010101) = 1101$
- $S_7(110011) = 0101$
- $S_8(110000) = 0000$

So the S-box outputs are:

0000 1100 0010 0001 0110 1101 0101 0000

f) Concatenate the results to get B

$B = 0C216D50$

g) Apply the permutation to get $P(B)$

Using the DES permutation P :

$P(B) = 921C209C$

h) Calculate $R_1 = P(B) \oplus L_0$

$$R_1 = 921C209C \oplus 33FF3300$$

$$\boxed{R_1 = A1E3139C}$$

Also, by Feistel structure:

$$L_1 = R_0 = 0F550F55$$

i) Write down the ciphertext

After one round, the block is:

$$L_1 R_1 = 0F550F55 A1E3139C$$

In DES, the block entering the final permutation is $R_1 L_1$:

$$R_1 L_1 = A1E3139C 0F550F55$$

Applying the final permutation IP^{-1} , the one-round DES ciphertext is:

$$\boxed{FE9CAB8927503251}$$

Final answers

a.

$$\boxed{K_1 = F4FD9864B65A}$$

b.

$$\boxed{L_0 = 33FF3300}, \boxed{R_0 = 0F550F55}$$

c.

$$\boxed{E(R_0) = 85EAAA85EAAA}$$

d.

$$A = 711732E15CF0$$

e.

$$0000\ 1100\ 0010\ 0001\ 0110\ 1101\ 0101\ 0000$$

f.

$$B = 0C216D50$$

g.

$$P(B) = 921C209C$$

h.

$$R_1 = A1E3139C$$

i.

- after one round, before final permutation:

$$0F550F55A1E3139C$$

- one-round DES ciphertext after final permutation:

$$FE9CAB8927503251$$